MILITARY SPECIFICATION

CONTROLLERS, STATIC, NAVAL SHIPBOARD

1. SCOPE

1.1 This specification covers static, class HI shockproof electric power controllers and associated devices for Naval shipboard use. The equipment covered by this specification utilizes semiconductor devices for power handling.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-S-901 - Shock Tests, H. I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-E-917 - Electric Power Equipment, Basic Requirements for (Naval Shipboard Use).
MIL-D-963 - Drawings, Electrical, Hull and Mechanical Equipment for Naval Shipboard Use.
MIL-C-2174 - Controllers, Direct-Current, Naval Shipboard.
MIL-C-2212 - Controllers, Alternating-Current, Naval Shipboard.
MIL-R-10509 - Resistors, Fixed Film (High Stability). General Specification For.
MIL-M-15071 - Manuals, Equipment and Systems.
MIL-P-15137 - Provisioning Technical Documentation For Repair Parts For Electrical and Mechanical Equipment (Naval Shipboard Use).
MIL-F-15160 - Fuses; Instrument, Power and Telephone.
MIL-C-17361 - Circuit Breakers, Air, Electric, Insulated Enclosure (Shipboard Use).
MIL-C-17588 - Circuit Breaker (Automatic - ALB) and Switch, Toggle (Circuit Breaker, Non-automatic - NLB), Air, Insulated Enclosure, 125 volts and Below AC or DC, Naval Shipboard.
MIL-R-18281 - Rectifiers, Metallic: Selenium, Copper Oxide, and Magnesium-Copper Sulfide Types, Naval Shipboard Use.
MIL-E-22843 - Equipment, Low Noise Level (Naval Shipboard Use).
MIL-R-27208 - Resistors, Variable, Wirewound (Lead Screw Actuated), General Specification For.

STANDARDS

MILITARY

MIL-STD-167 - Mechanical Vibrations of Shipboard Equipment.
MIL-STD-740 - Noise Measurements of Shipboard Machinery and Equipment.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules.

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(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, New York.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)
IC1 - Control, Industrial.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 155 East 44th Street, New York 17, New York).

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Qualification. - Controllers furnished under this specification shall be a product which has been tested and has passed the qualification tests specified herein, and has been listed on or approved for listing on the applicable qualified products list.

3.2 General requirements. - Controllers shall conform to MIL-E-917 and National Electrical Manufacturer's Association (NEMA) standards for Industrial Control. Manufacturer's shall serialize all complete controllers. This identification number shall appear on the identification plate (see 3.4.1.7.2). Whenever a conflict of requirements exists, the requirements of MIL-E-917 shall govern.

3.2.1 Painting. - All parts not fabricated of a corrosion resisting material or having a corrosion resistant treatment as specified in MIL-E-917, except for semiconductors, shall be painted as specified therein except that only one coat of gray enamel shall be applied. Touching-up will be permitted for marks or scratches due to factory handling.

3.3 Design. -

3.3.1 Design characteristics. - Static controllers shall have the following characteristics, as specified (see 6.1):

Voltage input:

Alternating current (a.c.)	Direct current (d.c.)
Special	
440 volts, three phase	250 nominal (175-355) volts
115 volts, single phase	230 volts
115 volts, three phase	115 volts

Type:

A.C. loads	D.C. loads
On-off	On-off
Adjustable voltage	Adjustable voltage
	Speed control by adjustable voltage
	Field control
	Armature control
	Armature and field control

Performance:
- Non-automatic
- Automatic

Protection:
- Undervoltage protection (LVP)
- Undervoltage release (LVR)

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Function:
Motor starting
Motor starting and speed adjustment or selection
Motor starting and reversing
Miscellaneous (for example, control for electrical heaters and lighting)

3.3.1.1 Master switches.- Master switches shall conform to MIL-C-2174 or MIL-C-2212 in location and type. Pressure and temperature transducers and static proximity limit switches will be permitted.

3.3.1.2 Controllers.- The following output characteristics shall apply to controllers:

Automatic adjustable voltage output (adjustable voltage only)
Zero volts to 90 percent or greater of input voltage, r.m.s. A means, with a locking device, shall be provided for setting a lower limit of voltage within the above voltage range. The setting shall be as specified (see 6.1).
Increments shall be in steps not greater than 2.5 percent of rated voltage output.
Repeatability shall be within plus or minus 3 percent of adjusted voltage output for any given input.
Regulation shall be plus or minus 2 percent of adjusted voltage output. If motor load, a.c. tachometer feedback may be used.
Voltage output (on-off only)
The voltage drop through the controller, when operating at maximum output voltage, shall be held to a mimimum. In no case shall the final output voltage be less than 90 percent of input voltage.
Power output (a.c. controllers only)
.6 PF lag to unity PF
Controllers shall be designed for their maximum current and power rating at the low power factor point and full output voltage.

3.3.2 Frequency.- A.C. equipment shall be designed for operation at a frequency of 60 or 400 cycles plus or minus 5 percent, as specified (see 6.1).

3.3.3 Duty and service.- Unless otherwise specified in the contract or order (see 6.1), equipment shall be designed for continuous duty and general-purpose service. If other duty or service is required, these shall be in accordance with NEMA standards for the specified duty or service (see 6.1).

3.3.4 Grounding.- Equipment shall be designed for operation on an ungrounded system but shall operate satisfactorily with an accidental ground on any power conductor. Current transformer secondaries shall be grounded to the controller enclosure.

3.3.5 Simplification and standardization.- Controllers shall be simplified and standardized to the maximum extent possible, consistent with obtaining reliable operation and the required operating features. Each manufacturer shall provide that on-off and adjustable voltage controllers have identical power handling sections for the same or range of horsepower loads. Replaceable logic modules will allow utilization in different applications. Encapsulated modules shall be approved by the Bureau of Ships.

3.3.5.1 Circuit arrangement.- A means of line disconnect, that isolates all loads from the line, shall be provided. Circuit breakers (without auto trip) or rotary switches rated to interrupt maximum line current will be satisfactory. Circuits shall be designed for fail safe operation, as practicable. That is, failure of protective features shall cause the load to be removed from the line. Single-phase brakes, used with three-phase motors, shall be connected to a different phase than the control circuit.

3.3.5.2 Motor starting.- All controllers for motor starting shall have logic modules which provides for ramp voltage output for soft starts. That is, on initial start-up the voltage shall rise at a constant volts per time unit or exponential curve from zero to maximum output voltage. The starting time shall be adjustable from 1 second to 50 seconds. Any derating of the controller necessary to accomodate the starting time shall be noted on the drawing. Controllers for loads other than motors need not provide for ramp voltage output on start-up. All main power semiconductor devices shall be protected from short circuit by suitable current limit fuses.

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3.3 5.3 Master switch markings.- Master switch markings and information plates shall be clear and concise. NEMA markings and arrangements shall be used where feasible. In other cases, the controller manufacturer's standard markings and arrangements shall be used to avoid the supplying of controllers which have the same characteristics but differ in master switch marking.

3.3.5.4 Indicating light colors.- Indicating lights shall be provided and shall have the following colors unless otherwise specified (see 6.1):

Normal operation: Green
Power available: White

3.3.6 Pilot circuit protection.- All control circuit leads from controllers to remote devices shall be protected against damage from short circuits. Leads which are in series with current limiting components located within the enclosure are considered to be adequately protected. All other leads to remote devices shall be protected by 10 ampere fuses located within the enclosure.

3.3.7 Overload protection.- Overload protection shall be provided for each motor during both starting and running conditions. The rating of overload relays shall fall within the range of approximately 115 to 125 percent of the motor nameplate current. Overload relays shall be of the hand reset type. Resetting shall be accomplished by means external to the enclosure. When two single-pole overload relays are used with three-phase motors, resetting of the two relays shall be accomplished by a single button. Temperature sensitive devices imbedded in the end turns of the motor may be used.

3.3.7.1 Emergency-run feature.- When specified (see 6.1), controllers shall be provided with a means of defeating the overload relay This feature may be combined with the means of resetting the overload relay or it may be obtained with a separate pushbutton. It shall require manual operation for the duration of the emergency and shall not be combined with the start pushbutton.

3.3.8 Ambient temperature.- Unless otherwise specified (see 6.1), equipment shall be designed for a maximum of 50° centigrade (C.) ambient.

3.3.9 Parts.-

3.3.9.1 Overload relays and inherent protectors.- Overload relays shall conform to MIL-C-2174, MIL-C-2212 or the applicable requirements of this specification. Inherent protectors will be permitted (for example, positive or negative coefficient thermistors imbedded in the end turns of the motor stator).

3.3.9.2 Resistors and rheostats.- Resistors and rheostats shall conform to MIL-E-917, MIL-R-10509 and MIL-R-27208 as applicable.

3.3.9.3 Fuses.- Fuses shall be used in series with power semiconductors and shall be of the type suitable for protection of semiconductor devices. All other fuses shall conform to style F61 of MIL-F-15160.

3.3.9.4 Terminals.- Terminal boards and studs shall conform to the applicable requirements of MIL-C-2212.

3.3.9.5 Switches and transducers.- Switches and transducers shall conform to MIL-C-2174, MIL-C-2212 or the applicable requirements of this specification.

3.3.9.6 Circuit breakers.- Circuit breakers shall conform to MIL-C-17361 and MIL-C-17588, types NQB or NLB.

3.3.10 Temperature rise.- Temperature rise of transformers, reactors, and so forth shall not exceed values based upon the limiting hot-spot temperature of 105° C. for class A, 130° C. for class B, 155° C. for class F, 180° C for class H and 200° C. for class N insulation. Semiconductor temperature rise shall conform to MIL-E-917. The temperature requirements of the referenced parts specifications shall be met, as applicable, for other individual parts.

3.3.11 Operational requirements.-

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3.3.11.1 Endurance. - Static controllers shall be capable of withstanding 20,000 cycles of operation under rated load at rated voltage and any power factor above 50 percent. Controllers for motor loads shall also be capable of withstanding a 200 hour endurance test with a motor load of the maximum controller rating.

3.3.11.1.1 Transducers and proximity limit switches. - Transducers and proximity limit switches shall be capable of withstanding 100,000 cycles of operation under rated load at rated voltage.

3.3.11.2 Vibration. - Equipment shall be designed to withstand type I vibration tests of MIL-STD-167 without mechanical damage or malfunctioning.

3.3.11.3 Shock. - Equipment shall withstand the shock tests specified in MIL-S-901 without the use of shock mounts. With rated voltage applied, equipment shall be designed so that the load shall not be energized or de-energized because of shock. All control devices shall function properly during and after the shock test.

3.3.11.4 Undervoltage. - Controllers shall be designed to interrupt the load between 80 and 70 percent of rated input voltage, unless otherwise specified (see 6.1). Controllers shall not interrupt the load due to transients of a 5 second duration or less.

3.3.11.5 Voltage limits. - Controllers shall be designed to operate satisfactorily between 80 and 110 percent of rated input voltage. The controllers need not maintain the requirements of 3.3.1.2 over the whole range. A particular percentage change in the input shall cause no greater change in output.

3.3.11.6 Noise. - Low noise equipment shall be supplied and be designed to meet the requirements of MIL-E-22845. Equipment shall meet grade C for airborne and structureborne noise, unless otherwise specified (see 6.1).

3.3.12 Insulation resistance. - Insulation resistance shall not be less than 10 megohms.

3.4 Enclosures. - Enclosures shall be of the following types, as specified (see 6.1):

(a) Dripproof.
(b) Watertight.
(c) Submersible (15 feet - 30 minutes).
(d) Explosionproof (class I, group D).

3.4.1 Detail requirements. - Enclosures shall conform to NEMA Publication ICl for the designated type and the requirements of 3.4.1.1 through 3.4.1.6 herein.

3.4.1.1 Dripproof. - Dripproof enclosures shall meet NEMA requirements when inclined to any angle not greater than 45 degrees from the normal position.

3.4.1.2 Access. - Enclosures, over 1-1/2 cubic feet and other than explosionproof, shall have hinged doors. Doors wider than 18 inches shall be removable. Removal must be readily accomplished with no more than 18 inches of free space in front of the enclosure. Doors on dripproof enclosures shall be secured by captive thumb screws or other fasteners with all parts captive. All other enclosures shall be secured by captive screws. Removable covers may be used on enclosures smaller than 1-1/2 cubic feet. If standard screws are used to secure such covers, the screws need not be captive.

3.4.1.3 Mounting details. - Unless enclosures are designed to carry the weight of the equipment enclosed, the equipment shall be mounted in such a manner that the weight is transmitted directly to the ships structure and not borne by the wall of the enclosure.

3.4.1.4 Cable entrance. - Enclosures shall be so designed that all necessary cables of adequate size can be brought into either the top or bottom. Dripproof, watertight and submersible enclosures, larger than 4 cubic feet, shall have removable cable entrance plates. Gaskets shall be used as required.

3.4.1.5 Moisture drain. - Watertight and submersible enclosures larger than 60 cubic inches shall have a moisture drain in the bottom. The opening shall be closed with a 1/4 inch or larger pipe plug.

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3.4.1.6 Marking.- All controllers and master switches shall be permanently and clearly marked as to function of each position subject to manual operation. Painting or stenciling will not be permitted.

3.4.1.7 Diagrams and identification plates.-

3.4.1.7.1 Connection diagram and table of overload heaters.- Each controller shall include a connection diagram, a schematic diagram, a description of operation and a table of overload heaters or coils. This information shall be printed by a process that is nonfading, protected by transparent plastic and secured to the enclosure or to the door or cover. The preferred method is to laminate the diagram between two layers of polyester film by a heat sealing process. The diagram is then cemented in place. Alternate methods are to laminate the diagram between sheets of vinyl chloride or to provide metal photoprocess diagrams. Such diagrams shall be secured by means of metal clips or rivets.

3.4.1.7.2 Identification and information plates.- Each equipment shall be provided with an identification plate containing the manufacturer's name and catalog number. Controller identification plates shall also contain the voltage, maximum horsepower rating, and serial number (see 3.2). The preferred location is inside the enclosure. Identification plates for relays and contactors may be adhesive backed metal not less than 0.010 inch in thickness. Other identification or information plates shall be not less than .030 inch in thickness. Plates shall be brass, corrosion-resistant steel or anodized aluminum marked by etching, engraving, stamping or photoprocess. Etching, engraving or stamping shall not be less than 0.003 inch deep.

3.4.1.7.3 Warning plates.- Controllers energized from more than one source of power shall be provided with a plate to warn that both power sources must be disconnected before working on the controller. The word "WARNING" shall be in prominent red letters. The material and method of marking shall conform to 3.4.1.7.2. The plate shall be mounted on the front of the enclosure.

3.5 Repair parts.-

3.5.1 Interchangeability.- Each repair part shall be equal to the original and suitable for replacement without special tools, and or without the removal and replacement of rivets or pins dependent upon a force fit. Hexagon keys for socket head screws, or standard size screwdriver bits for Phillips head screws, are not considered to be special tools. When one form of a part or device will serve as a replacement for other forms, without affecting interchangeability or impairing the serviceability of the controller, this form may be furnished as a repair part on the same basis as if all forms were identical. Special tools are defined as those tools not listed in the Federal Supply Catalog. (Copies of this catalog may be consulted in the office of the Government inspector.)

3.5.2 Quantity of repair parts.-

3.5.2.1 Controller repair parts.- Repair parts, furnished with controllers, shall be furnished in sets as specified in table I. The number of sets shall be as specified (see 6.1).

Table I - Controller repair parts.

Parts	Factor code	Quantity per set					
		Number of components					
		1	2	3	4	5-8	9-20
Capacitors	F	1	1	1	2	2	4
Encapsulated items	G	1	1	1	1	2	3
Overload heaters	D	1	2	3	3	4	6
Pushbutton elements	G	1	1	1	1	2	3
Rectifiers	E	1	1	2	2	3	4
Resistors	F	1	1	1	2	2	4
Rheostats	F	1	1	1	2	2	4
Silicon controlled rectifiers	D	1	2	3	3	4	6

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3.5.2.1.1 Standard items.- Repair parts should not be furnished for Navy standard items, such as fuses, indicator lights and switches. Repair parts shall be furnished for proprietary items.

3.5.2.2 Repair parts lists.- Submission of repair parts lists to the Inventory Control Point for stock numbering action shall be in accordance with MIL-P-15137.

3.6 Drawings.- Drawings shall conform to class B and C of MIL-D-963. The multisheet format is allowed. Drawings shall be prepared by the controller manufacturer and contain the following minimum data, as applicable:

3.6.1 Controllers.-

(a) List of material, listing the enclosure and all components. Each item, other than Navy standard, should be identified by the catalog number assigned by the company responsible for the design of the item. Navy standard items shall be identified by the Navy drawing or type number.
(b) Outline of the front and top views of the controller, giving dimensions and essential details of the enclosure, including quantity and size of mounting bolts required. The center of gravity shall be shown in both views for controllers over 100 pounds.
(c) Connection and schematic diagrams of the control circuit, including connections to master switches and other remote devices. The rating of all fuses shall be indicated. A legend table shall be included.
(d) Description of operation of the controller.
(e) Description of adjustments which can be made on the equipment, such as the range of time for a timing relay.
(f) Table of overload heater or coil catalog numbers, with motor full load current ranges.
(g) Current transformer data.
(h) Voltage and volt-ampere rating of control circuit transformers.
(i) Table of autotransformer or reactor data, containing turns, wire size and covering, coil insulation and impregnation and rating of taps.
(j) Table of applicable resistor catalog numbers with corresponding step-by-step and overall resistances, continuous ampere ratings of each step and identification as to where used.
(k) Capacitor data, including type (paper or electrolytic) and microfarad and voltage rating.
(l) Semiconductor data including rating and complete ID for ordering from original supplier (VEDEC number is adequate), and the following:
    (1) Rectifiers using silicone diodes.
        a. Circuit symbols and piece number.
        b. Maximum reverse voltage peak and r.m.s. (complete rectifier circuit).
        c. Maximum load current d.c. amperes (complete circuit).
        d. Maximum diode case temperature at design ambient-use for diodes rated against core temperature.
        e. Maximum ambient temperature directly adjacent to diodes—use for diodes rated against ambient temperature—no cooling plates.
        f. Cooling plate size, thickness and material where so mounted.
        g. Air velocity if forced cooled (linear feet per minute).
        h. Circuit (for example, single phase bridge, and so forth).
        i. Total diodes in circuit.
        j. Diodes in series per arm.
        k. Diodes in parallel per arm.
        l. Description of insulation hardware if insulated from a cooling plate.
    (2) Rectifiers using selenium cells (in stacks or cartridges).
        a. Circuit symbol(s) and piece number.
        b. Maximum a.c. voltage applied to the rectifier circuit.
        c. Maximum d.c. load current through rectifier circuit.
        d. Maximum cell temperature (shall not exceed 75° C).
        e. Cell dimensions (3 by 3 inches, 5 by 6 inches, and so forth).
        f. Air velocity if forced cooled (linear feet per minute).
        g. Rectifier circuit (single phase bridge, and so forth).
        h. Total stacks or cartridges used in circuit.
        i. Stack or cartridge circuit (if different from rectifier circuit).
        j. Cells in series per arm of stack or cartridge.

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k. Cells in parallel.
l. Special requirements specified in the purchase order but not covered in MIL-R-18281 but considered necessary for this application (such as a tighter reverse current limit for magnetic amplifier or control application).

(3) Voltage regulator diodes.
a. Circuit symbol(s) and piece number.
b. Range of actual voltage levels which circuit will accept (with or without circuit adjustment from regulator diodes and still give required equipment performance____ volts to____volts.
c. Regulation (including temperature error) permitted in an installed regulator diode under all conditions of operation (assume equipment is initially adjusted for optimum performance)± ___ millivolts.
d. Maximum level of steady-state average reverse current through regulator diode (steady-state shall be considered any time in excess of 30 seconds): ____ ma.
e. Maximum transient reverse current under any operating condition____ ma for___ sec. duration and_____ frequency of occurrence.
f. Maximum ambient temperature at diode (no cooling plate).
g. Maximum core temperature of diode (complete for diodes rated against case temperature).
h. Cooling plate size, thickness and material where used.
i. Air velocity if forced cooled (linear feet per minute).
j. Total diodes in circuit.
k. Diodes in series per circuit element (complete where more than one circuit element is used in the voltage regulator such as in a bridge type regulator circuit).
l. Description of insulation hardware if insulated from a cooling plate.

(4) Voltage reference diodes (see regulator diodes also).
a. Circuit symbol(s) and piece number.
b. Range of reference voltage levels which circuit will accept (with or without circuit adjustment) from the voltage reference diode(s): ___volts to ___volts.
c. Error permitted in an installed voltage reference diode under all conditions of operation: ± ___ millivolts.
d. Circuit design current through voltage reference.
1. Nominal or average value:___ ma.
2. Regulation about nominal value: ±___ ma.
e. Maximum ambient temperature (include internal heat rise above outside design ambient): ___ a.
f. Number of voltage reference elements in series: (Operation in parallel is not permitted.)

(5) Transistors.
a. Circuit symbol(s) and piece number.
b. Maximum collector to emitter voltage___ volts.
c. Maximum collector to base voltage___ volts.
d. Maximum collector current___ma.
e. Maximum base current___ma.
f. Maximum collector power dissipation___ mw.
g. Maximum ambient temperature (include internal heat rise at transistor at 50°C, design ambient ___°C.
h. Maximum core temperature of transistor at 50°C. equipment design ambient and any condition of required equipment operation ___°C. (Where rating is based on case temperature.)
i. Cooling plate size, thickness and material where so mounted.
j. Description of insulating hardware if insulated from a cooling plate.

(6) Silicon controlled rectifiers.
a. Similar information to transistor data, as applicable.
b. List of exceptions to this specification.
c. Descriptive data of the equipment which shall include rating, degree of enclosure and catalog number. Include derating due to starting characteristics (see 3.3.5.2).

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d. Actual weight of the equipment and an unboxed set of repair parts.
e. List of repair parts with piece numbers for each item, the name of each item, the quantity of each item recommended for one set of repair parts and the manufacturer's identification for the item. This list shall conform to 3.5.2.
f. Table of revisions including a brief description of each revision of the drawing.
g. Title block.

3.6.2 Master switches. -
(a) Data required by items (a) through (m) through (r) of 3.6.1.
(b) Outline, showing two major views, giving dimensions and essential details of the enclosure.
(c) Switch symbol or diagram.

3.6.3 Certification data. - Certification data shall include the following:
(a) Navy contract or shipbuilder's order and item number.
(b) Controller manufacturer's order and item numbers.
(c) The master drawing numbers of the controller and all associated master switches.
(d) The catalog numbers of the controller and all associated master switches.
(e) Applicable specifications (including issue, amendment and date).
(f) Voltage, horsepower and ampere rating of motor.
(g) Drawing number of motor and other devices associated with the controller.
(h) Number and identification of ships involved.
(i) Application on ship.
(j) Quantity per ship.
(k) List of repair parts (if repair parts are furnished on a per set basis).
(l) Title block.

3.6.4 Applicability. - These drawing requirements are applicable to initial procurements of controllers, master switches and repair parts. They are not intended to be applicable to orders for repair parts purchased separately.

3.7 Qualification under reference specifications. - When any specification, which forms a part of this specification, requires qualification tests, only products which have been approved for listing on the applicable Qualified Products List, on the date of invitation for bids or date of initiation of purchase action, shall be utilized. In the event no Qualified Products List has been issued, the supplier shall request instructions as to what testing will be required to determine the product meets the requirements of the specification.

3.8 Manuals. - Manuals shall be in accordance with type I of MIL-M-15071 and the following:
(a) Include photographs of controller with door both opened and closed.
(b) Block diagram showing main circuit components. Description of operation relating to block diagram.
(c) Corrective maintenance shall include tables and diagram showing points of measurement of voltage, current, resistance, and so forth.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Tabulation of inspection. - Qualification, comparison and quality conformance inspection shall be as shown in table II.

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Table II - Inspection agenda.

Description and order of inspection	Qualification inspection	Comparison inspection	Quality conformance inspection	Requirement paragraph	Inspection paragraph
Workmanship, fit and materials	X	X	X	3.2	4.6.1
Weight	X	X	-	-	4.7.1
Creepage and clearance	X	X	-	-	4.7.2
General operation	X	X	X	3.3.1 and 3.3.5.2	4.7.3
Output wave shape	X	-	-	-	4.7.11.1
Noise	X	-	X	3.3.11.6	4.7.4
Voltage variation	X	X	-	3.3.11.4 and 3.3.11.5	4.7.5
Temperature rise	X	X	X	3.3.10	4.7.6
Endurance	X	-	-	3.3.11.1	4.7.11
Vibration	X	-	-	3.3.11.2	4.7.7
Insulation resistance	X	X	-	3.3.12	4.7.8
Dielectric	X	X	X	-	4.7.9
Shock	X	X	-	3.3.11.3	4.7.10
Explosionproof	X	X	-	3.4	4.7.12.1 and 4.7.12.2
Enclosure-workmanship, fit, materials, and finish	X	X	X	3.4	4.7.12.4
Effectiveness of enclosure	X	X	-	3.4	4.7.12.5
Enclosure shock	X	X	-	3.3.11.3	4.7.12.6

4.3 Qualification inspection 1/.- Qualification inspection shall be conducted at a laboratory satisfactory to the Bureau of Ships. Qualification inspection shall consist of the examination and tests specified in table II. The manufacturer shall request authorization of such tests from the Bureau of Ships. A proposed test agenda shall be forwarded with the request for Bureau of Ships approval.

4.3.1 Samples for qualification.- Two samples each of the basic controller types specified in 3.3.1 shall be submitted for test. Otherwise, tests on two adjustable voltage types will cover all a.c. types and tests on two armature and field control types will cover all d.c. types. The samples shall be rated at the maximum horsepower offered. Three equipment samples, of other than complete controllers, shall be submitted for their applicable qualification tests. Class C drawings shall be submitted with the request for authorization to test covering types of controllers for which approval is desired.

1/ Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.3 and 6.3.1).

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4.4 Comparison inspection.- One year after qualification inspection and each year in which the equipment is offered for delivery thereafter, one equipment of each design shall be subjected to the comparison inspection specified in table II. These tests are not required to be conducted on more than one out of fifty equipments of a given design manufactured over a period of three years. Three copies of the test report shall be forwarded to the Bureau of Ships.

4.5 Quality conformance inspection.- Each equipment shall be subjected to the quality conformance inspection specified in table II.

4.6 Inspection.-

4.6.1 Equipment.- Each equipment shall be given an examination to determine that the design, material, and workmanship conform to this specification. The fit of parts shall be observed, with particular reference to interchangeability of replaceable parts. Conformance to approved drawings with respect to material, finish, construction, dimensions and weight, shall be verified. This examination should not include disassembly to the point that performance, durability or appearance would be affected.

4.6.2 Repair parts.- All repair parts shall be subjected to an examination to ascertain that the materials, workmanship and finish are first class in every respect and that they conform fully to the manufacturer's approved drawings. The principal object of this inspection shall be to determine that the repair parts are equal to or better than the original parts. If the inspector has reason to doubt the ready interchangeability of the repair parts with the original parts, he may require a demonstration of such interchangeability.

4.6.3 Additional inspection.- Where other specifications form a part of this specification, examination and tests shall be conducted as required in the referenced specification.

4.7 Test procedures.-

4.7.1 Weight.- The weight of the equipment, and the weight of the repair parts, shall be taken and recorded.

4.7.2 Creepage and clearance.- Creepage and clearance distances shall be demonstrated by actual measurement to be in accordance with MIL-E-917.

4.7.3 General operation.- General operation shall include tests necessary to ascertain that the master switches operate as ordered and that the required operating features have been met. That is, the pushbutton or selector switch shall be operated to see if the controller energizes and de-energizes the load as required. If an automatic master switch is ordered, it shall be cycled to see if the controller output follows as required. The test shall also include a check of the ramp voltage start feature of motor controllers. For qualification and quality conformance inspection only, the controller shall be tested in a 50 °C. ambient for 48 hours with a load of maximum current rating. Equipment shall not malfunction. Alternate methods of testing for early failure of semiconductor devices will be considered.

4.7.3.1 Adjustable voltage controllers.- Controllers shall be tested with a load of controller's maximum rating to prove that the controller meets the requirements of 3.3.1.2. This test shall be conducted for qualification and quality conformance inspection only. The test shall include the following checks:

(a) Vary the signal input and observe whether the output voltage appears stepless. If not stepless, record the step value in percentage of rated voltage output.
(b) Set the signal input and record it and the output voltage. Change the signal input at least 25 percent of setting and return to recorded value of input. The output shall again be recorded. Repeat at least five times. This check shall begin with the controller at room ambient. One reading shall be taken each minute after energization.
(c) Set the signal input at five different positions and leave in those positions for at least one minute. Record the voltage output continuously at each setting with the maximum rated load on the controller.

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4.7.4 Noise.- Noise tests shall be conducted in accordance with the procedure for solidly mounted equipment of relatively light construction as specified in MIL-E-22843. All devices shall be energized simulating actual operation. Short test reports as defined in MIL-STD-740 shall be furnished (see 3.3.11.6). Only one controller of each type need be tested for each order. The test shall be conducted at both half and full load for a.c. and d.c. adjustable voltage controllers. On-off controllers shall be tested at full load only.

4.7.5 Voltage variation.- The controller shall be tested to determine conformance with 3.3.11.4 and 3.3.11.5. The controller shall be connected to a suitable voltage supply that can be varied from 70 percent to 110 percent of the controller rated input voltage. It shall be determined that the controller will not malfunction due to the input voltage varying between 80 and 110 percent of rated value. The master switches shall be operated at the low and high points to determine that the controller operates satisfactorily. The voltage shall then be decreased to 70 percent to determine that the controller de-energizes the load between 70 and 80 percent of rated input voltage. Failure to meet these requirements shall be cause for rejection.

4.7.6 Temperature rise.- The test methods to be employed and the precautions to be observed in conducting temperature rise tests shall be in accordance with MIL-E-917 and the following:

4.7.6.1 General.- Temperature rise tests shall be made under conditons equivalent to normal operating conditions at rated input voltage and frequency and the duty specified, to determine that the rises specified in 3.3.10 are not exceeded. Tests shall also be conducted at the low speed and 60 percent rated speed, if specified (see 6.1). Temperature rise tests shall be made on individual parts of the completely assembled equipment including semiconductors, resistors, reactors, capacitors, and so forth. The specified voltage shall be applied to all parts simultaneously so that the mutual heating effect of the several parts will be present. Care shall be taken to assure enough readings are taken to monitor the worse cases and be indicative of the hottest components.

4.7.6.2 Duration of tests.- The temperature rise test shall be continued until the temperature rises that can be observed during the test have attained a steady final value. That is, where the final reading differs from the previous reading by less than 1 percent over a period of 5 minutes.

4.7.7 Vibration.- Equipment shall be subjected to the type I vibration test of MIL-STD-167. The equipment shall be mounted in normal operating position. Operation shall be checked at each frequency of vibration.

4.7.7.1 Definition of failure.-

(a) Breakage or appreciable distortion of any parts, including mounting bolts.
(b) Electrical malfunctioning. All design characteristics shall continue to be met. Protective features shall not malfunction or become inoperative.
(c) Low insulation resistance. After vibration test, the insulation resistance shall conform to 4.7.8.
(d) Failure to pass inspection. The equipment shall be disassembled following checks (a) through (c) and inspected thoroughly for damage. The extent of disassembly need be only to the point where the condition of the equipment can be easily observed. Any observed effects of the vibration should be recorded.

4.7.8 Insulation resistance.- Insulation resistance shall be as specified in 3.3.12. The measurement shall be made with an insulation-resistance-indicating meter with a full scale reading of 100 or 200 megohms, an open circuit voltage of 50 volts and a voltage of at least 45 volts across a resistance of 1 megohm. The temperature and relative humidity shall be measured and recorded. Insulation resistance measurements shall be corrected to 25 °C. Corrections shall be made on the basis of insulation resistance doubling for each 15 °C. decrease in temperature. No correction shall be made for humidity.

4.7.9 Dielectric.- Dielectric tests shall be conducted in accordance with NEMA Publication IC1-2.42. The tests shall be made upon the equipment with all semiconductors shorted after completion of all other tests except the shock test.

4.7.10 Shock.- Shock tests shall be conducted in accordance with MIL-S-901 for grade A, class I equipment as specified in 4.7.10.1 and 4.7.10.2.

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4.7.10.1 Method of mounting.- For assembled controllers and for other equipment mounted separately on the ship's structure, the fixture 4A mounting of MIL-S-901 shall be used. For individual controller devices, the smallest panel, fixture 6E of MIL-S-901, on which the device can be mounted shall be used. Equipment shall be mounted on the shock machine in a manner simulating normal shipboard mounting. Equipment shall be tested both energized and de-energized if contactors and relays are used. Otherwise, the test is required only with the equipment energized.

4.7.10.2 Definition of failure.-

(a) Breakage or appreciable distortion of any parts, including mounting bolts.
(b) Electrical malfunctioning. The conditions of 3.3.11.3 shall apply. Protective features shall not malfunction or become inoperative.
(c) Low insulation resistance. After shock tests, the insulation resistance shall conform to 4.7.8.
(d) Low dielectric strength. After shock tests, the dielectric test shall be conducted at a voltage equal to 65 percent of the voltage required by 4.7.9.
(e) Failure to pass inspection. The equipment shall be disassembled following checks (a) through (d) and inspected thoroughly for damage. The extent of disassembly need be only to the point where the condition of the equipment can be easily observed. Any observed effect of the shock should be recorded.

4.7.10.3 Disposition.- If a controller satisfactorily passes shock test, it may be furnished as an item under a contract or order after correction of any minor damage.

4.7.11 Endurance.- The endurance test shall be conducted in accordance with 3.3.11.1. A cycle of operation shall consist of 3 minutes energized and 2 minutes de-energized. This cycle shall be repeated 20,000 times. Forced draft cooling shall be used during each de-energized period. Salt spray shall be introduced into the controller during at least 500 periods that it is de-energized. The following data shall be recorded during test:

(a) Temperature rises of power semiconductors to monitor change in thermal impedance. (Record at least every 500 cycles and any significant points. Readings shall be taken at the end of the energized period.)
(b) Input and output current and voltage waveshapes also showing magnitude. (Recorded at time of temperature readings.)
(c) Condition of terminal points and metal parts (denote corrosion, loosening of connections, and so forth. Record at time of other readings).

4.7.11.1 Motor controllers.- If approval is desired for motor loads a 200 hour test is required with an actual motor load of the maximum horsepower rating of the controller. For an automatic controller, a control input shall be required to cycle simulating possible operating conditions. The control input shall cycle 5 times during each hour of operation. One cycle shall be a change of the input signal from minimum to maximum and back to minimum. The test shall be divided into four 50-hour periods. There shall be 25 starts and stops during each of these periods. Temperature rise data shall be taken on motor stator and controller parts. Controller input and output voltage and current shall be recorded. An oscillogram of motor current and voltage and their wave shapes shall be taken during the first and last start in each test period. The above test procedure will suffice to approve both nonautomatic and automatic controllers. If continuous testing is not practiced, the minimum testing period shall be 7 hours at any one time.

4.7.11.2 Transducers and proximity limit switches.- The endurance test shall be conducted in accordance with 3.3.11.1.1. The 100,000 cycles shall be conducted at rated load. A cycle shall consist of the output signal of the tested device changing from minimum or off to maximum or on to minimum or off.

4.7.12 Enclosures.-

4.7.12.1 Qualification tests.- Qualification tests are required only on explosionproof enclosures and enclosures included with complete equipments submitted for qualification approval. Explosionproof enclosures shall be tested by the Bureau of Mines.

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4.7.12.2 Comparison inspection.- Comparison inspection shall be conducted on a representative sample of each group of similarly constructed enclosures at intervals of 3 years. All enclosures in a group shall be similarly constructed and have the same features as to doors, louvers and degree of enclosure. The volume of the largest enclosure in the group shall not exceed four times the volume of the smallest.

4.7.12.3 Quality conformance inspection.- Quality conformance inspection shall be made on each enclosure.

4.7.12.4 Workmanship, fit, materials and finish.- Enclosures shall be inspected to determine that the workmanship, fit, materials and finish are first class in every respect.

4.7.12.5 Effectiveness of enclosure.- Enclosures shall be inspected to determine compliance with 3.4.1 and as specified in 4.7.12.5.1, 4.7.12.5.2, and 4.7.12.5.3.

4.7.12.5.1 Dripproof.- No tests are required. An examination shall be made to determine that falling moisture or dirt will not interfere with the mechanical or electrical operation of the equipment enclosed when inclined to any angle not exceeding 45 degrees from the vertical.

4.7.12.5.2 Watertight.- At the option of the manufacturer, the test may consist of submerging the equipment to a depth of 3 feet for 5 minutes in lieu of the NEMA hose test. Equivalent pressure or vacuum tests, with the enclosure submerged, are acceptable provided that the resulting pressure is on the outside of the enclosure.

4.7.12.5.3 Gaskets and seals.- Before conducting tests, all gasketed joints and seals (cable entrance plates excluded) shall be broken and reassembled three times.

4.7.12.6 Shock.- Shock tests of enclosures shall be conducted under the conditions of 4.7.10.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation and for storage of onboard repair parts.-

5.1.1 Static controllers.-

5.1.1.1 Preservation and packaging.- Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation and may conform to the supplier's commercial practice when such meets these requirements.

5.1.1.2 Packing.- Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the supplier's commercial practice when such meets these requirements.

5.1.1.3 Marking.- Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the supplier's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, manufacturer's name and destination.

5.1.2 Onboard repair parts.- Onboard repair parts shall be packaged in heat sealable, laminated paper and foil bags or in heated sealable plastic film. Shipping containers and methods of packing shall comply with the carrier's regulations applicable to the mode of transportation. Interior and exterior shipping containers shall be marked with the Federal stock number.

6. NOTES

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6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Design characteristics (see 3.3.1).
(c) Voltage setting (see 3.3.1.2).
(d) Frequency (see 3.3.2).
(e) Duty and service, if other than continuous and general-purpose (see 3.3.3).
(f) Indicating light colors (see 3.3.5.4).
(g) Whether emergency-run feature is required (see 3.3.7.1).
(h) Ambient temperature, if other than 50°C. (see 3.3.8).
(i) Undervoltage protection (see 3.3.11.4).
(j) Grade of noise required, if other than grade C (see 3.3.11.6).
(k) Type of enclosure (see 3.4).
(l) Number of repair part sets required (see 3.5.2.1).
(m) Load characteristics and requirements (horsepower or wattage rating, speed range, torque requirements, overhauling loads, and so forth).
(n) Control input, if automatic controller is ordered.
(o) If adjustable voltage controller is being ordered, require temperature rise test on motor and controller together at applicable load, low speed condition and 60 percent rated speed (see 4.7.6.1).

6.2 Controllers for both a.c. and d.c. loads may be ordered in accordance with this specification. Care should be exercised to assure proper characteristics are listed. Application of adjustable voltage a.c. controllers under this specification shall be limited to variable torque loads (for example, centrifugal pumps) unless specifically approved by the Bureau of Ships.

6.3 With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in Qualified Products List QPL-23994, whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the Bureau of Ships, Department of the Navy, Washington, D.C. 20360, and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.3.1).

6.3.1 Copies of "Provisions Governing Qualification" may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

Preparing activity:
Navy - Ships
(Project 6110-N067Sh)

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES" IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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